A Preliminary Study on the Multitemporal Analysis of Cryosphere Radar Sounder Data

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ABSTRACT

Radar Sounder (RS) data contain information on subsurface geology and are analyzed mostly with automatic techniques on single-pass acquisitions. A few preliminary studies on multitemporal RS data acquired on the cryosphere focus on possible advances in ad-hoc data acquisition strategy and ice-sheet monitoring, such as the percolation zone. However, challenges related to data corregistration and the inherent characteristics of the target hinder the multitemporal analysis. This paper analyzes bi-temporal radargrams with partially overlapping footprints (thus showing information from neighboring geographical areas) in the cryosphere and defines a strategy to estimate candidate changes. The paper proposes projecting and locally corregistering the radargram pairs at different depths to identify the expected changes due to glacier displacement and snow accumulation. Comparing the corregistered radargrams, we identify candidate unexpected changes in the ice sheet morphology that glaciologists should further validate. The proposed method is validated on several radargram pairs acquired by MCoRDS-3 in Antarctica in 2014 and 2016.

Keywords: Multitemporal analysis, Cryosphere, Image processing, Radar sounder, Subsurface sensing.

1. INTRODUCTION

Radar Sounders (RSs) are active sensors that probe the cryosphere subsurface by acquiring profile cross-sections called radargrams. Radargrams contain information on subsurface geology that is analyzed with automatic techniques, including deep learning for semantic segmentation[1,2] and classical machine learning for target detection, such as subglacial lakes[3] and buried cavities.[4] Automatic techniques proposed in the literature extract precise information about subsurface geology, mostly from single acquisitions, without exploiting the multitemporal information available in repeated acquisitions. This is because there are few multitemporal cryosphere RS campaigns with fully or partially overlapping footprints, given that the main purpose of sounding campaigns was to produce gridded subglacial topographic maps.[5] Further, even campaigns planned to acquire over similar tracks may have different latitudes and longitudes, given the non-ideality of the acquisition process and aircraft motion. Hence, tracks are intertwined, and we can define three cases (froubsurface geology, mostly from the left in figure 1) – i) the tracks intersect, and thus most of the RS footprints overlap, and the radargrams image the same along track area. ii) The tracks are distant, the footprints do not overlap, and radargrams do not share the same information. iii) Finally, the tracks are not too distant, and the footprint partially overlaps so that radargrams share similar information. Moreover, available RS data are generally acquired with varying instrument parameters and configurations, resulting in distinct geophysical observables[6,7] because subsurface scattering is strongly frequency-dependent. However, multitemporal analyses are fundamental for better understanding the cryosphere evolution and monitoring in the context of climate change.

A few preliminary studies on multitemporal RS data analysis focus on possible advances in ad-hoc multitemporal data acquisition strategy[5] and the analysis of the percolation zone.[8] Schroeder et al.[5] analyzed the challenges related to the multitemporal analysis of RS data and suggested a strategy for having more multitemporal orbital sounding. 8 analyzed repeated Operation Ice Bridge (OIB) accumulation radar acquisitions of the Greenland firn with coherent change detection for identifying percolation and refreezing in porous. The

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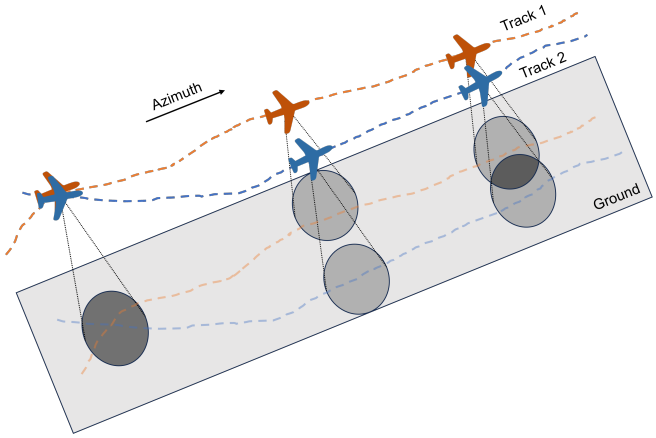


Figure 1. Example of realistic acquisition geometry. From the left: i) the tracks intersect, and thus, most of the RS footprints overlap, and the radargrams image the same along the track area. ii) The tracks are distant, the footprints do not overlap, and radargrams do not share the same information. iii) Finally, the tracks are not too distant, and the footprint partially overlaps so that radargrams share similar information.

work suggested that the most critical elements for the multitemporal analysis of subsurface radar data are the corregistration and motion compensation methods. However, challenges related to the data corregistration and the inherent characteristics of the targets hinder the multitemporal analysis.

This paper analyses bi-temporal RS acquisitions with partially overlapping footprints in the cryosphere and defines a strategy to estimate expected and unexpected candidate changes. Expected changes are related to the glacier sliding in the along-track direction, snow accumulation and ablation, and changes in the medium propagation properties (e.g., temperature regime) in the range direction. Ice stratigraphy changes as morphological shape variations in both along-track and range directions. To identify the changes in the vertical direction, we compensate radargrams for the aircraft fluctuations and locally corregister them in the range direction using the bedrock as a reference, assuming it is invariable in the considered period. We estimate the ice displacement between radargrams using a patch-based crosscorrelation to identify the along-track changes and refine those in the range. The proposed method provided promising results on several pairs of radargrams acquired in Antarctica in 2014 and 2016, spanning the same track.

The remaining part of the paper has the following structure: Section 2 proposes the novel method, Section 3 describes the dataset, and Section 4 presents the experimental results. Finally, Section 5 concludes the paper and presents the future works.

2. PROPOSED METHOD

Let us consider two radargrams $\mathbf{R}_{t_1}$ and $\mathbf{R}_{t_2}$ with partially overlapping footprints acquired in two different moments t_1 and t_2. The radargrams image a similar portion of the surface and subsurface. Each radargram can be defined as a 2D matrix of N_T^i traces (columns) and N_S^i samples (rows):

$$\mathbf{R}_i = \{R_i(r,a) | a \in [1, N_T^i], r \in [1, N_S^i]\}, \tag{1}$$

where $i = [t_1, t_2]$, R_i is the power of the reflections in the radargrams, and (a, r) are the coordinates in an azimuth/range reference system, respectively. Each trace $a \in [1, N_T^i]$ in the radargrams can also works a longitude/latitude reference system by a latitude $x(a)$ and longitude $y(a)$. Hence, the radargrams can be formally written as $\mathbf{R}_i = \{R_i(r, x(a), y(a)\}$ where $x(a) \in [-90; +90]$ and $y(a) \in [-180; +180]$. Adjacent traces of the same radargram capture information on the adjacent vertical portions of the subsurface and image adjacent geologic targets.

The proposed methodology consists of four steps (see Figure 2)– preprocessing, radargram projection, radargram local corregistration, and change detection. The preprocessing adapts $\mathbf{R}_{t_1}$ and $\mathbf{R}_{t_2}$ so that they span over

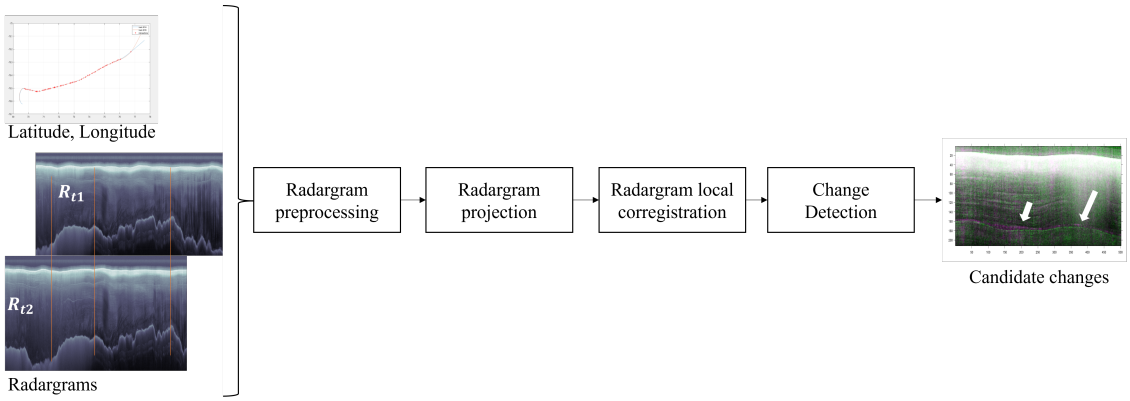


Figure 2. Block scheme of the proposed method for the multi-temporal analysis of radargrams.

the same coordinates (i.e., latitude and longitude) and have footprints that partially or completely overlap, i.e., less distant than the antenna footprint (see Figure 1). Secondly, we project $\mathbf{R}_{t_1}$ onto $\mathbf{R}_{t_2}$ to force $N_T^{t_1} = N_T^{t_2}$ and $N_S^{t_1} = N_S^{t_2}$. Thirdly, we locally corregister the radargrams to compensate for the ice sheet's vertical mass balance change and horizontal motion. Finally, we perform change detection over pairs of radargrams.

2.1 Radargram preprocessing

The radargram preprocessing identifies the rangelines of the two radargrams that span over a similar latitude and longitude. To this end, we analyze the latitude and longitude of the tracks of $\mathbf{R}_{t_1}$ and $\mathbf{R}_{t_2}$ to identify the starting and ending values of latitude and longitude that are shared and we remove the other rangelines. Moreover, since the tracks follow the acquisition geometry in Figure 1, we calculate the average distance of the tracks and remove the rangelines that are more distant than the antenna footprint so that radargrams image a similar subsurface area. The output is two radargrams acquired over the same area but not the same number of columns so that $N_T^{t_1} \neq N_T^{t_2}$ (see Figure 5).

2.2 Radargram projection

To force $N_T^{t_1} = N_T^{t_2}$, we project the longest radargram onto the shortest one using dynamic time warping (DWT).[9] DWT performs the optimal matching of the t_1 and t_2 tracks considering the two sequences of latitude X_{t_1}, X_{t_2} and longitude Y_{t_1}, Y_{t_2}. The matching is done considering the following rules– i) every value of (X_{t_1}, Y_{t_1}) must be matched with those of (X_{t_2}, Y_{t_2}) and vice versa. ii) The first value of (X_{t_1}, Y_{t_1}) must be matched with the first value of (X_{t_2}, Y_{t_2})(but it does not have to be its only match). iii) The last value of (X_{t_1}, Y_{t_1}) must be matched with the last value of (X_{t_2}, Y_{t_2}) (but it does not have to be its only match). iv) The mapping of (X_{t_1}, Y_{t_1}) to (X_{t_2}, Y_{t_2}) must be monotonically increasing, and vice versa, i.e., if $x_{t_1} > x'_{t_1}, (x_{t_1}, x'_{t_1}) \in X_{t_1}$ and $y_{t_1} > y'_{t_1}, (y_{t_1}, y'_{t_1}) \in Y_{t_1}$ then there must not be two values $x_{t_2} > x'_{t_2}, (x_{t_2}, x'_{t_2}) \in X_{t_2}$ and $y_{t_2} > y'_{t_2}, (y_{t_2}, y'_{t_2}) \in Y_{t_2}$, such that x'_{t_1} and y'_{t_1} are matched with x_{t_2} and y_{t_2} respectively, and x_{t_1} and y_{t_1} are matched with x'_{t_2} and y'_{t_2} respectively, and vice versa. v) Finally, we force the latitude and longitude of the intersection points between the two sequences to match. The optimal match is the match that satisfies all the rules and has the minimal cost, where the cost is computed as the sum of absolute differences for each matched pair of latitude and longitude values.

Once the latitude and longitude indices of the rangelines of $\mathbf{R}_{t_1}$ that should be projected are identified, we compute the matched latitude and longitude sequence and the matched radargram. The output of this step is a set of comparable radargrams with $N_T^{t_1} = N_T^{t_2}$.

2.3 Radargram local corregistration

A couple of radargrams spanning the same geographical area may contain expected changes due to the ice sheet and the platform horizontal and vertical displacements.[8] These displacements should be mitigated to detect unexpected changes (e.g., layer disruption).

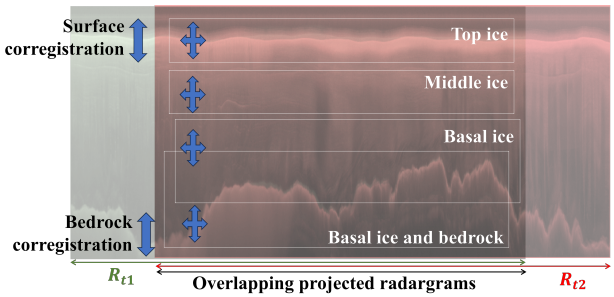


Figure 3. Crosscorrelation geometry of radargrams $\mathbf{R}_{t_1}$ in green and $\mathbf{R}_{t_2}$ in red– the white boxes indicate the four sub-radargrams that are cross-correlated separately, and blue arrows indicate the crosscorrelation directions.

The horizontal displacement is due to the ice sheet sliding toward the coast. Hence, radargrams that were acquired in two different time stamps and having a track perpendicular to the ice sheet sliding direction image two different portions of the ice sheet that, however, are expected to be similar. Radargrams that are acquired in two different time stamps and have a track parallel to the ice sheet sliding direction image the same portion of the ice sheet but shifted forward or backward. Horizontal changes can be mitigated by cross-correlating the radargrams and identifying the two parts of the radargrams that image the same portion of the ice sheet.

Vertical (i.e., range direction) offsets are due to platform fluctuations, systemic differences in the radar settings between acquisitions, changes in the mass balance of the ice sheets over the years,[8] and changes in the dielectric properties of the ice sheet. Changes in the mass balance are related to the accumulation and ablation processes of the snow and ice over the years. Platform fluctuations are mitigated as done in the literature[1,3] by considering the aircraft's vertical coordinate. We perform coarse corregistration in the range direction to isolate these changes in the two date radargrams.

The vertical dimension of radargrams $\mathbf{R}_{t_1}(r,a)$ and $\mathbf{R}_{t_2}(r,a)$ is different given that it depends on the dielectric properties and thermal regime of the ice sheet in t_1 and t_2, and the different acquisition geometry. Moreover, the sliding velocity of the ice varies with the depth. Hence, we horizontally divide the radargrams into four separate stripes: top ice, middle ice, basal ice, and basal ice and bedrock (see Figure 3). The upper sub-radargram (called top ice) images the surface and the firn subjected to the generation of ice slabs and water percolation.[8] The middle radargram (middle ice) images the ice stratigraphy, the sub-radargram below images the basal area, and the last sub-radargram images the basal interface. The four stripes are cross-correlated separately to perform a local correlation that considers the morphological characteristics of the ice structure. The corregistration of two radargram stripes $\mathbf{R}^s_{t_1}(r,a)$ and $\mathbf{R}^s_{t_2}(r,a)$ is done by considering the cross-correlation index in 10:

$$\gamma(r',a') = \frac{\sum_{(r,a)} \left[\mathbf{R}^s_{t_1}(r,a) - \overline{\mathbf{R}^s_{t_1}}(r',a')\right] \left[\mathbf{R}^s_{t_2}(r-r',a-a') - \overline{\mathbf{R}^s_{t_2}}\right]}{\sqrt{\sum_{(r,a)} \left[\mathbf{R}^s_{t_1}(r,a) - \overline{\mathbf{R}^s_{t_1}}(r',a')\right]^2 \sum_{(r,a)} \left[\mathbf{R}^s_{t_2}(r-r',a-a') - \overline{\mathbf{R}^s_{t_2}}\right]^2}} \tag{2}$$

2.4 Change detection

Once we obtain the two cross-correlated radargrams $\mathbf{R}^c_{t_1}(r,a)$ and $\mathbf{R}^c_{t_2}(r,a)$, we can analyze them to identify candidate changes in the ice pack. Here, we apply a log ratio function to highlight candidate changes as large differences in the radargram regions, thus with a high probability of change. The log-ratio is defined as

$$\mathcal{L}(a,r) = \log \frac{\mathbf{R}^c_{t_1}(r,a)}{\mathbf{R}^c_{t_2}(r,a)}. \tag{3}$$

The log ratio is a well-known measure from the SAR literature that enhances the contrast between changed and unchanged areas in a pair of radar images, making it easier to identify changes. This metric effectively highlights changes because it normalizes the radar data, reducing the impact of noise (speckle) and other artifacts. Finally, the log-ratio image $\mathcal{L}(a,r)$ is expected to have small values in areas with no changes and high values in areas with large changes. Hence, we thresholded $\mathcal{L}(a,r)$ to identify candidate changes related to the ice sheet morphology.

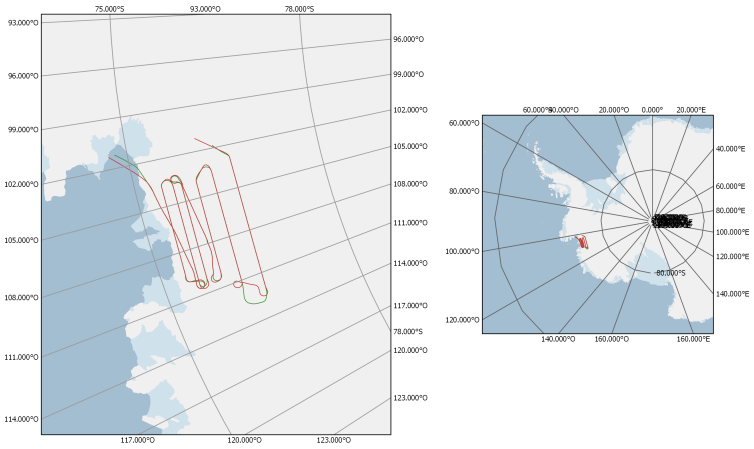


Figure 4. Ground tracks of the campaigns 20141103-04 in red and 20161103-06 in green.

Parameters	**Antarctica dataset**
Central frequency (MHz)	190
Wavelength (m)	1.58
Bandwidth (Mhz)	50
Sampling Frequency (MHz)	50
Transmitted Power (kW)	6
Altitude AGL (km)	0.5
Range Resolution in ice (m)	2.6
Azimuth Resolution (m)	27.5
Ground sampling distance (m)	15

Table 1. Parameters of MCoRDS-3[11] for the campaigns acquired in Antarctica in 2014 and 2016. Geometrical resolutions of the radargrams, considering the ice dielectric permittivity equal to $\varepsilon_r = 3.15$. AGL stands for above ground level

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3. DATASET DESCRIPTION

To assess the effectiveness of the proposed method, we consider two campaigns sharing the same track in Antarctica by the MCoRDS-3 instrument[11] from the Center for Remote Sensing of Ice Sheets (CReSIS). We choose campaigns 20141103-04 and 20161103-06 that were acquired in 2014 and 2016, respectively, and share most of the track (see Figure 4).

The data are preprocessed for each campaign, including range compression, azimuth focusing with Synthetic Aperture Radar (SAR), and partial clutter compensation, leading to the spatial resolutions shown in Table 3. The preprocessing also involves concatenating adjacent radargrams and correcting for fluctuations caused by undesired movements during acquisition, such as pitch, roll, and yaw, which might have compromised signal quality. As described in,[1] we normalize the radargram power based on the surface power for each column to mitigate power fading due to the aircraft rolling movement on curves. We determined the maximum power for each azimuth column using the vector of the range coordinate of the surface $s(a)$. The surface power $P_s(a)$ for each azimuth trace was defined as $P_s(a) = \mathbf{R}(a, s(a))$. We masked the region above the surface to focus on the subsurface.

Finally, we compared the tracks of the two campaigns to select the traces that share the same coordinates (i.e., span over the same tracks). We identified eight pairs of radargrams; however, we only reported the analysis for 4 (see Table 2 for the details).

Couple ID	Radargram ID start	Radargram ID end	Number of traces	Number of samples
RDR1	20141103_04_002	20141103_04_006	16661	2500
	20161103_06_002	20161103_06_006	16666	2500
RDR2	20141103_04_008	20141103_04_010	9996	2500
	20161103_06_008	20161103_06_010	9999	2500
RDR3	20141103_04_012	20141103_04_014	9996	2500
	20161103_06_012	20161103_06_014	9999	2500
RDR4	20141103_04_016	20141103_04_019	13328	2500
	20161103_06_015	20161103_06_019	16666	2500

Table 2. Details of the pairs of radargrams analyzed.

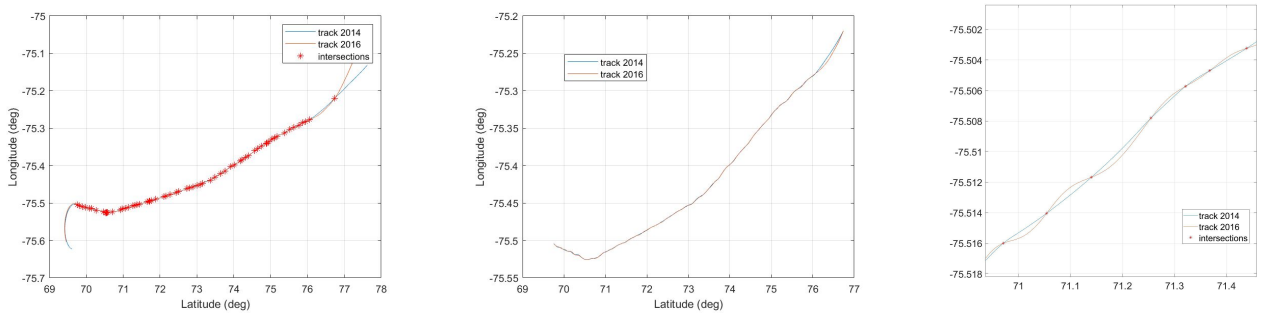


Figure 5. Example of the tracks (plot of the latitude vs longitude) of the pair RDR1 with the 2014 track in blue and the 2016 track in orange– on the left are the original tracks with highlighted intersections (red stars), the central plot shows the overlapping tracks after the processing, and the plot on the right is a zoom of the left plot.

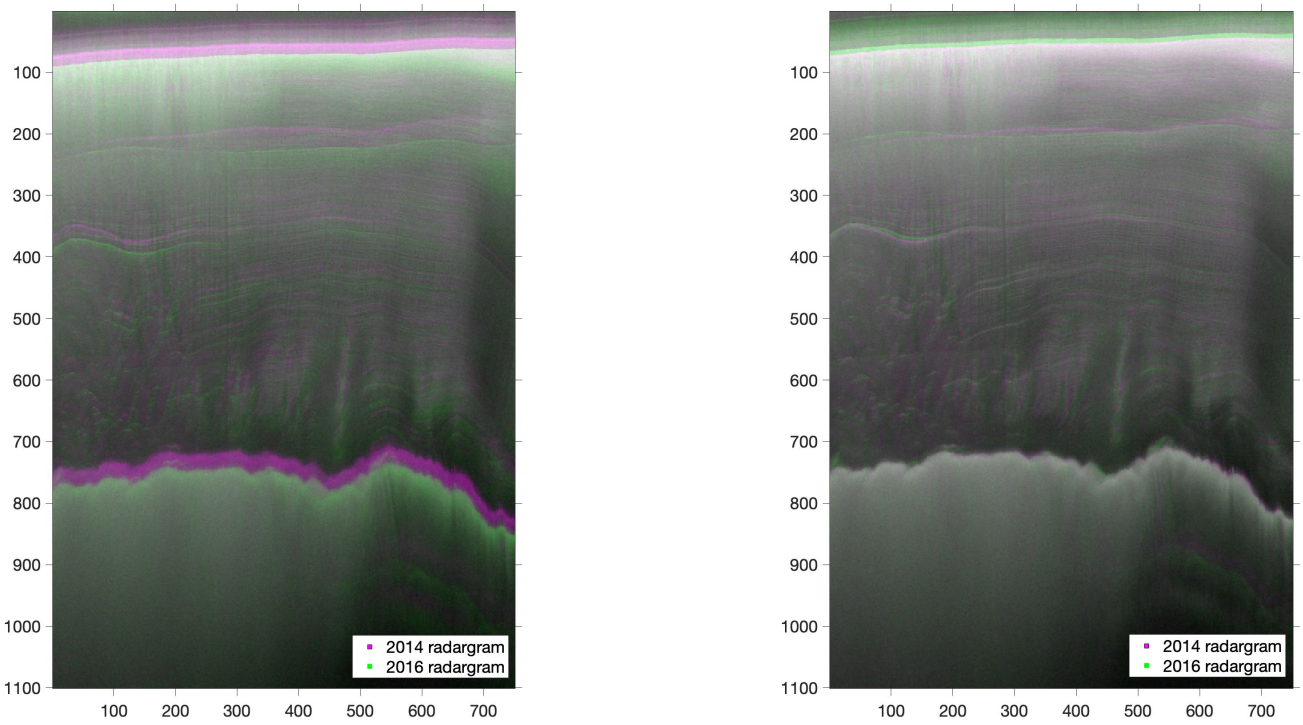


Figure 6. False color composition of the radargrams (R: radargram of 2014, G: radargram of 2016)– on the left the input radargram and on the right the corregistered ones.

Couple ID	Top ice	Middle ice	Basal ice
RDR1	3	0	1
RDR2	8	1	1
RDR3	1	0	0
RDR4	3	2	1

Table 3. Horizontal displacement with respect to the basal ice and bedrock in pixels.

Couple ID	Top ice	Middle ice	Basal ice
RDR1	8	5	1
RDR2	9	7	4
RDR3	9	6	3
RDR4	9	6	2

Table 4. Vertical displacement with respect to the basal ice and bedrock in pixels.

4. EXPERIMENTAL RESULTS

The results of the radargram projection are shown in Figure 5– on the left, there is the plot of the latitude vs. longitude of the 2014 (in blue) and 2016 (in orange) tracks before the preprocessing and the projection. The red stars indicate the intersections between the latitude and longitude tracks. The tracks and the radargrams are cut so that the starting and ending coordinates of the radargrams are the same. Then, we project the 2014 radargram onto the 2016 one to have the same number of columns in the azimuth direction. Figure 6 shows the resulting radargram on the left. Figure 6 on the right shows the two radargrams corregistered with respect to the bedrock, which is expected to not change in time.

Figure 7 shows the zooms of the false color composition of the four areas corregistered separately. Tables 3 and 4 report the estimated horizontal and vertical displacements between the 2014 and 2016 radargrams. These displacements are expected given the ice sheet changes (see Section 2.3). The top ice sub-radargram has a greater vertical displacement than the other sub-radargrams. We argue that the displacement is expected and is due to large temperature variations between 2014 and 2016 or the changes in mass balance. We analyzed the fluctuations of the aircraft to prove that they are not responsible for the difference in ice thickness. The lateral aircraft movement is not correlated with the larger number of rows. The horizontal displacement corresponds to a high corregistration value (0.96 on average) for the surface, which is higher than in other portions of the radargram. This is also expected as the ice velocity in the upper part of the ice sheet is generally larger than in the deeper ice sheet in the case of a frozen bed (as this is the case). In the middle ice, the corregistration brings a layering structure coincident (they appear white), and vertical streaks are observed. There is a very high correlation (0.92 on average) with a smaller horizontal displacement than the surface (5 pixels at the surface, 2 pixels in the middle section on average). The vertical offset (5 pixels at the surface, -2 pixels on average) indicates no accumulation or temperature variations in the middle layers. The correlation in the basal ice sub-radargrams is lower than in the previous cases. This is because there is a significant amount of noise (a large EFZ and minimal layering, which serves as a reference for the corregistration method); thus, displacement values could be more meaningful. The horizontal displacement values are similar to those of the middle section. The vertical displacement is on average of 2 pixels. Vertical lines continue to be visible, possibly due to percolation. The correlation (0.95 on average) in the bedrock sub-radargram is higher than in the basal area, given the presence of a sharp and strong reflector (bedrock). However, the tables 3 and 4 show that the offset is not constant between the experiments. We argue that the bedrock shows several processing artifacts and off-nadir clutter, making overlapping bedrock reflections appear very differently.

Finally, the false-color images (Figures 6 and 7) show expected and candidate changes related to the morphology of the ice layers. Candidate changes are mostly identified in the upper area of the ice sheet as it is the most dynamic part of the ice sheet. The basal ice (Figure 7.C) shows several changes in morphology and radiometry that are more ambiguous and may be due to differences in the acquisition geometry. Figure 7.D shows minor variations in the bedrock, which is expected to remain unchanged over time. The differences suggest slight discrepancies in the RS footprints that may result in false-positive changes. The false-color composition zoom in Figure 8 shows morphological differences in the upper part of the ice stratigraphy. The image shows a strong horizontal reflector (ice layer) with a different shape between the 2014 and 2016 radargrams. We highlighted

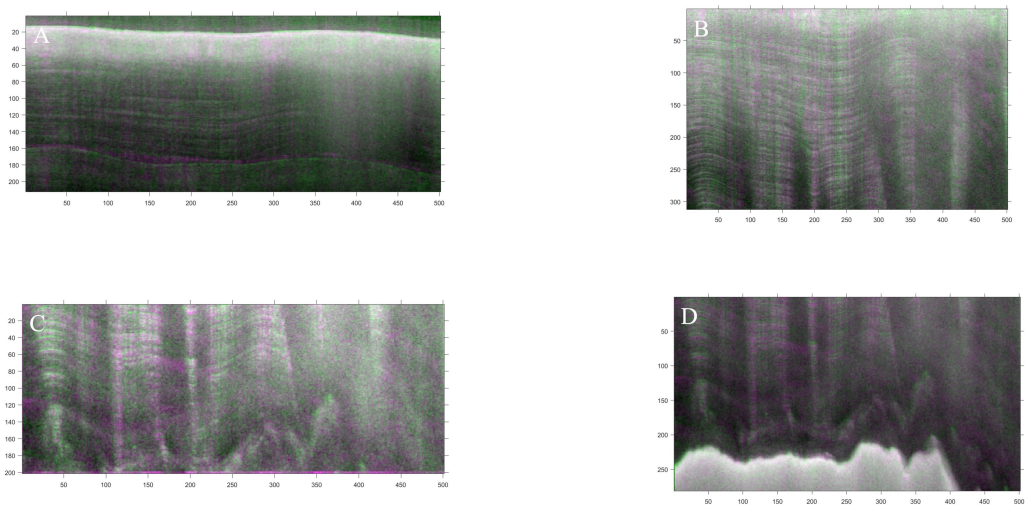


Figure 7. Zooms of the false color composition of the radargrams (R: radargram of 2014, G: radargram of 2016)– upper left top ice (surface), upper right middle ice (ice stratigraphy), bottom left basal ice, and bottom right the bedrock and basal ice.

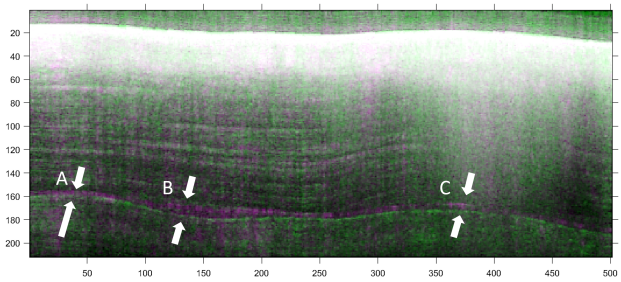


Figure 8. Zoom on the upper part of the ice sheet of the false color composition that shows morphological differences in the ice stratigraphy. The image shows a strong horizontal reflector with a different shape between the two radargrams. We highlighted with arrows three areas where the layer has a similar shape (A and C) and a different one (B).

three areas with arrows where the strong horizontal reflector has a similar shape (A and C) in the 2014 and 2016 radargrams and a different one (B). Figure 9 shows on the left the log-ratio image $\mathcal{L}$ for the top ice area in Figure 8 and on the right the change map. Both plots show strong changes close to the surface due to the water percolation, as expected from.[8] The second large change is in the morphology of the ice layer, which we also identified in the false color composition.

5. CONCLUSIONS AND FUTURE WORKS

This paper analyzes multitemporal radargrams acquired in the cryosphere and describes the preliminary results of the corregistration and identification of candidate changes in the ice sheet morphology. The corregistration results show that the horizontal displacement relative to the basal ice and bedrock corregistration is greater in the upper ice layers, with relatively consistent values observed in both the middle and basal ice. Notably, the upper ice layer exhibits significant variability, likely influenced by differences between the ice divide and coastal ice areas. In contrast, vertical displacement shows a clear gradient, higher in the upper ice, decreasing in the middle ice, and reaching its lowest values in the basal ice. Moreover, we identified candidate changes in the ice sheet morphology that glaciologists should further analyze.

In future work, we plan to consider the radargram radiometric values to perform the projection, allowing for a more in-depth analysis that does not only focus on the morphological elements. Moreover, we plan to analyze

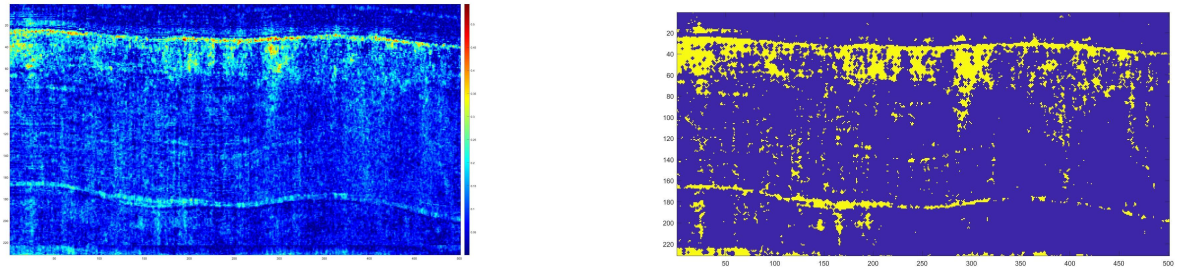


Figure 9. Examples of the change detection step results– the left Figure is the top ice log-ratio map $\mathcal{L}$. On the right, the threshold image highlights the candidate changes in yellow, and no changed area is shown in blue.

candidate changes in parallel acquisition pairs to validate candidate changes further and reconstruct a 3D map of the ice sheet and related changes in the acquisition pair.

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REFERENCES

[1] Donini, E., Bovolo, F., and Bruzzone, L., "A deep learning architecture for semantic segmentation of radar sounder data," *IEEE Transactions on Geoscience and Remote Sensing* **60**, 1–14 (2021).

[2] García, M. H., Donini, E., and Bovolo, F., "A weakly supervised transfer learning approach for radar sounder data segmentation," *IEEE Transactions on Geoscience and Remote Sensing* **61**, 1–18 (2023).

[3] Ilisei, A.-M., Khodadadzadeh, M., Ferro, A., and Bruzzone, L., "An automatic method for subglacial lake detection in ice sheet radar sounder data," *IEEE Transactions on Geoscience and Remote Sensing* **57**(6), 3252–3270 (2018).

[4] Donini, E., Carrer, L., Gerekos, C., Bruzzone, L., and Bovolo, F., "An unsupervised fuzzy system for the automatic detection of candidate lava tubes in radar sounder data," *IEEE Transactions on Geoscience and Remote Sensing* **60**, 1–19 (2021).

[5] Schroeder, D. M., "Pathways to multitemporal radar sounding in terrestrial glaciology," in [*IGARSS 2020-2020 IEEE International Geoscience and Remote Sensing Symposium*], 3731–3734, IEEE (2020).

[6] Winter, A., Steinhage, D., Arnold, E. J., Blankenship, D. D., Cavitte, M. G., Corr, H. F., Paden, J. D., Urbini, S., Young, D. A., and Eisen, O., "Comparison of measurements from different radio-echo sounding systems and synchronization with the ice core at dome c, antarctica," *The Cryosphere* **11**(1), 653–668 (2017).

[7] Chan, K., Grima, C., Rutishauser, A., Young, D. A., Culberg, R., and Blankenship, D. D., "Spatial characterization of near-surface structure and meltwater runoff conditions across devon ice cap from dual-frequency radar reflectivity," *The Cryosphere Discussions* **2022**, 1–22 (2022).

[8] Culberg, R. and Michaelides, R. J., "Towards coherent change detection for ice sheet near-surface process studies with airborne ice-penetrating radar," in [*IGARSS 2023-2023 IEEE International Geoscience and Remote Sensing Symposium*], 7800–7803, IEEE (2023).

[9] Müller, M., "Dynamic time warping," *Information retrieval for music and motion* , 69–84 (2007).

[10] Lewis, J. P., "Fast normalized cross-correlation," in [*Vision interface*], **10**(1), 120–123 (1995).

[11] Rodriguez-Morales, F., Gogineni, S., Leuschen, C. J., Paden, J. D., Li, J., Lewis, C. C., Panzer, B., Alvestegui, D. G.-G., Patel, A., Byers, K., et al., "Advanced multifrequency radar instrumentation for polar research," *IEEE Transactions on Geoscience and Remote Sensing* **52**(5), 2824–2842 (2013).